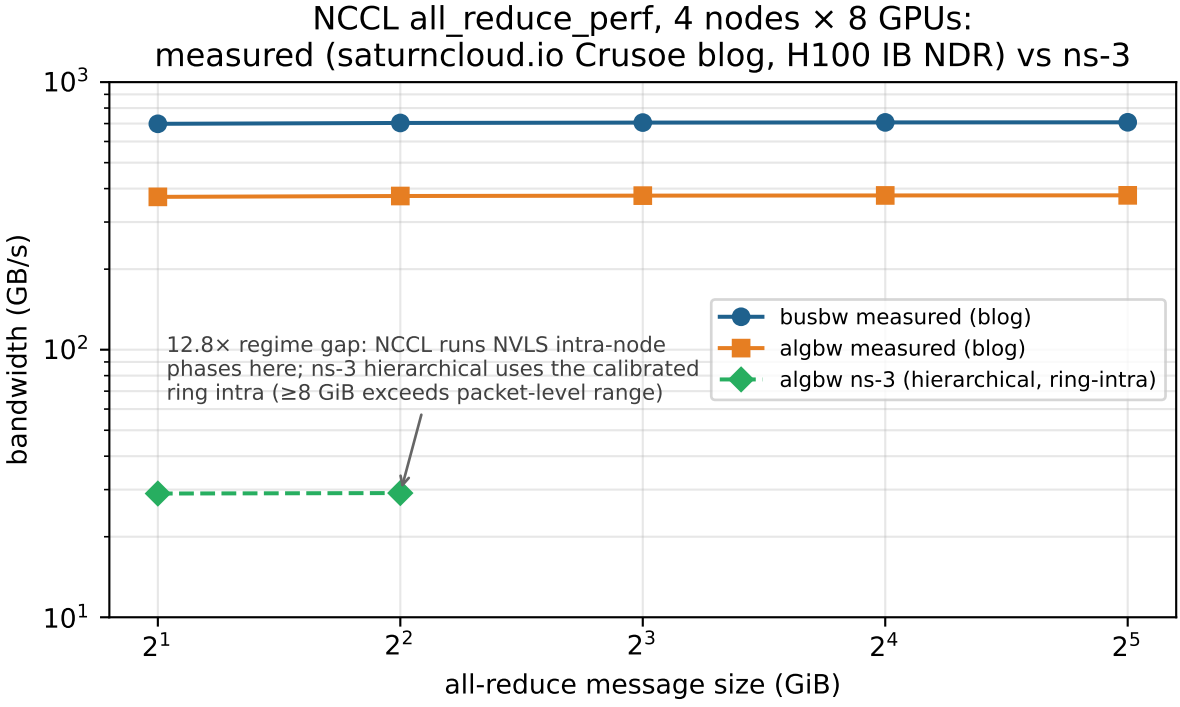
NCCL all_reduce_perf, 4 nodes × 8 GPUs:
measured (saturncloud.io Crusoe blog, H100 IB NDR) vs ns-3
bandwidth (GB/s)
10^3
10^2
10^1
busbw measured (blog)
algbw measured (blog)
algbw ns-3 (hierarchical, ring-intra)
12.8× regime gap: NCCL runs NVLS intra-node
phases here; ns-3 hierarchical uses the calibrated
ring intra (≥8 GiB exceeds packet-level range)
2^1
2^2
2^3
2^4
2^5
all-reduce message size (GiB)